

3 **(long)** The air-standard Atkinson cycle models a reciprocating engine where the isentropic compression (1→2) and constant-volume heat addition (2→3) processes are identical to the air-standard Otto cycle. Work is produced during isentropic expansion (3→4), where the expansion ratio $r_e = V_4/V_3$ from volumes V_3 to V_4 is larger than the compression ratio $r_c = V_1/V_2$ from volumes V_1 to V_2 . The expansion ratio results in the final pressure $p_4 = p_1$ such that the heat release is isobaric. Assume that the cycle fluid is air which can be treated as a perfect gas with constant specific heat capacity c_v and ratio of heat capacities γ .

(a) Sketch representative $p - V$ and $T - s$ diagrams for the air-standard Atkinson cycle. [5]

(b) Show that the temperature after heat addition is $T_3 = T_1 r_c^{\gamma-1} + q_{2-3}/c_v$, where T_1 is the initial temperature and q_{2-3} is the heat input per unit mass of air flow. [6]

(c) Show that the ratio $r = r_e/r_c$ for expansion such that $p_4 = p_1$ is

$$r = \left(1 + \frac{q_{2-3}}{T_1 c_v r_c^{\gamma-1}} \right)^{1/\gamma}.$$

[8]

(d) A given air-standard Atkinson engine has $V_1 = 0.3 \text{ m}^3$ and $r_c = 8$. The conditions prior to compression are temperature $T_1 = 300 \text{ K}$ and pressure $p_1 = 1 \text{ bar}$. The heat input per unit mass of air flow is $q_{2-3} = 1 \text{ MJ kg}^{-1}$. The air properties are $\gamma = 1.34$, gas constant $R = 287 \text{ J kg}^{-1} \text{ K}^{-1}$ and specific heat capacity at constant volume $c_v = 834 \text{ J kg}^{-1} \text{ K}^{-1}$.

(i) Calculate the temperatures after heat addition T_3 and after expansion T_4 . [5]

(ii) Calculate the efficiency of the engine. [6]